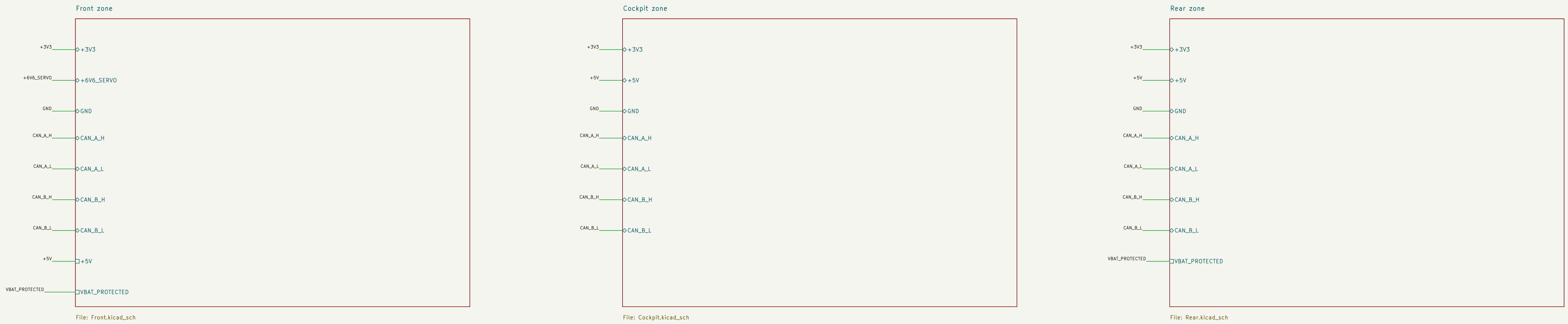
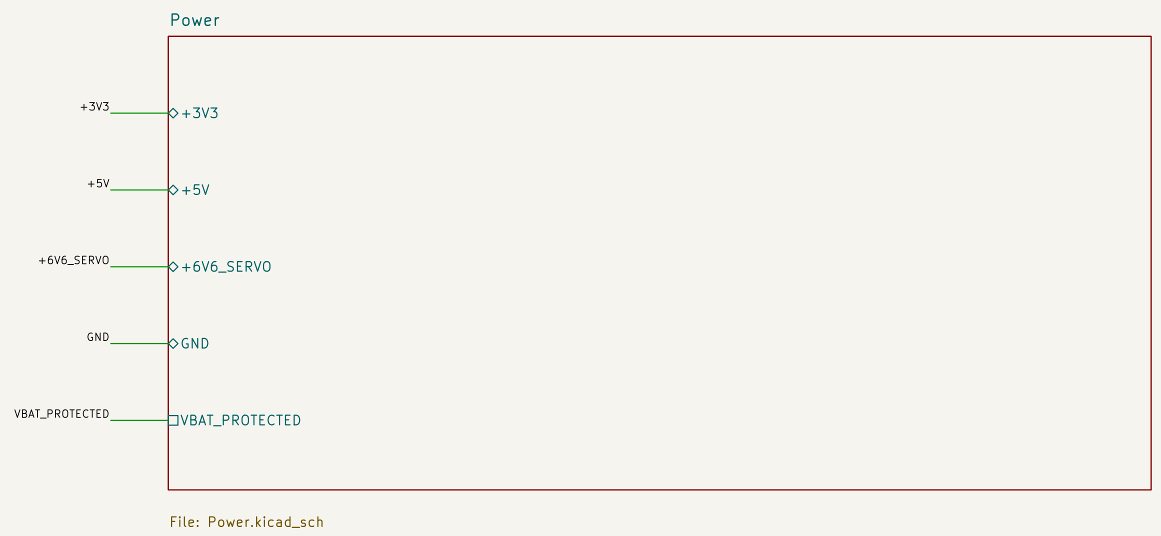
# LAB 4 — CONNECTIVITY OVERVIEW

Front and rear are the terminated ends. Cockpit taps both backbones.
Each zone has its own two isolation groups; termination remains on the bus side.
Actuator connectors / motor protection / physical layout remain unfinished.





CAN_A_H/L and CAN_B_H/L are separate root nets. Same-name local labels on this root join the shown sheet pins. Child labels connect only through those ports.

Power flags document external battery power and outputs after passive filters; they do not prove current capacity, regulation, thermal behavior or PCB completeness.

See Verification.txt and Design-review.csv for current checks and open decisions.

Shared nets: two CAN backbones, power and ground only.
PRELIMINARY — incomplete board; not for fabrication
**Team 4**
Sheet: /
File: Lab4.kicad_sch
***Title: Lab 4 — three-zone controller***
Size: A2 Date: 2026-10-20 **Rev: 0.3**
KiCad E.D.A. 10.0.4 Id: 1/5

# Front zone – CAN and servo current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

SERVO SHUNT – calculated proposal (not measured)

6.6 V regulator -> 10 mOhm shunt -> servo +
Servo GND returns directly to power ground.
Kelvin sense: separate thin traces from shunt pads to IN+/IN-.
Gain 50: Vout = 0.5 V/A. Proposed range 0–5 A -> 0–2.5 V.
12-bit / 3.3 V: 1.61 mA/count (ideal quantization).
At 5 A: shunt drop 50 mV; loss 0.25 W. Select >=1 W.
RC: 4.7k + 100n -> 339 Hz; proposed ADC 4 ksample/s.
Near-zero amplifier output clips; calibrate offset and gain.
This unidirectional circuit cannot measure regenerated current.
No servo idle/stall/noise measurements yet.
5 A measurement range is NOT a 5 A driver or regulator rating.

LAB3 header follows handout p17 top-to-bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

Servo and lamp circuits are below. Connector cable adapters still need physical verification.

STM32F401RET6 – independent HSI/PLL, no ST-LINK clock dependency

VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

TEST: active low on PC13, 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

SERVO AND VEHICLE LAMP OUTPUTS

Servo supply passes through R1. PWM uses PA9 and retains its reset pulldown.
J90 uses a new rated 3.96mm connector, not the existing servo plug.
Adapter must preserve GND / supply / signal; confirm the servo accepts 3.3V PWM.

3S LiPo BATTERY MONITOR: protected pack rail / 11 -> PA3 ADC1_IN3.
12.6V full -> 1.145V nominal ADC; 15V ceiling -> 1.364V.
Use calibrated VDDA, 480-cycle sample at <=21MHz; allow >=10ms filter settling.
No internal pullup/pulldown. Enable BOR; do not treat the MCU VBAT backup pin as pack input.
Firmware must inhibit drive at startup, stale/invalid samples and low battery.
This is a pack-voltage monitor, not per-cell protection or a hardware disconnect.

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY – incomplete board; not for fabrication
**Team 4**
Sheet: /Front zone/
File: Front.kicad_sch
***Title: Front zone – integrated logic and interfaces***
Size: A0
Date: 2026-10-23
**Rev: 0.3**
KiCad E.D.A. 10.0.4
Id: 2/5

## Rear zone — CAN and motor current sensing

CAN A: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

CAN B: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

LEFT MOTOR: L298 output -> ACS712 -> motor lead

RIGHT MOTOR: L298 output -> ACS712 -> motor lead

LAB3 header follows handout p17 top-to-bottom. Two GND pins first.
SRV/PWM_OUT probe actual actuator PWM; timing markers remain separate.
Physical header position / accessibility will be checked during PCB layout.

L298 and output circuits are below. Current limiting and motor operating envelope remains unresolved; heatsink geometry is specified; no motor-power sign-off.

Each motor: +/-5 A; VADC = 1.25 V + 0.0925 V/A * I.
Ideal 12-bit scale: 8.71 mA/count at VREF=3.3 V.
10k/10k divider keeps a 5 V sensor output at 2.5 V.
Filter source resistance: (10k || 10k) + 1k = 6k.
6k * 100n -> 265 Hz. Proposed ADC: 4 ksample/s/channel.
Stop motors before zero calibration; preserve signed readings.
Hall sensor has internal conductor, not an external shunt.
At 5 A and typical 1.2 mOhm, conductor loss = 30 mW.
Motor return lead connects to the OTHER L298 output, not GND.

## STM32F401RET6 — independent HSI/PLL, no ST-LINK clock dependency

VCAP1 is the INTERNAL regulator output. Only its local capacitor connects here.
Keep every 100n beside its VDD pin; keep the VCAP loop short. DS10086 Fig.2 / Fig.18.

SWD: pin1 VTref senses target voltage; do not feed a second 3.3V supply.
UART header is TX/RX/GND, not RS-232. NRST pulls low to reset.
BOOT0 low boots flash; fit recovery shunt only for bootloader recovery.

TEST: active low on PC13, 10k/100n = 1ms release RC.
Firmware must debounce and meet the 10ms local-fault deadline.
Reset and self-test are separate buttons.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

ENCODER INPUTS: board pins 1=4.0V regulated, 2=GND, 3=A, 4=B. Verify actual cable order.
10k pullups support open-collector outputs; push-pull input must stay <=5.5V.
U81 at 3.3V gives Schmitt hysteresis and MCU-safe outputs; unused inputs grounded.
Both A and B invert. Check direction/counts after the PCB firmware port.

## MOTOR POWER STAGE — cooling / stall behavior to verify

OUT1/OUT3 -> ACS712 IP+ -> IP- -> motor lead. Other lead -> OUT2/OUT4, NEVER ground.
ENA/ENB high + equal direction inputs gives dynamic braking. Enable low gives coast.
Existing 10k pulldowns keep enables low during reset. SENSE_A/B return to power ground.
This schematic has no autonomous motor-current limiter. A fuse and filtered ADC are not substitutes.
Keep the course L298N; check cooling and course stall-test expectations. 3.2A is a motor stall specification, not normal current.
The eight diodes clamp motor-terminal nets after the ACS712s. Keep loops short. Some PWM recirculation bypasses the sensors.

ENCODER SUPPLY: 5.1V -> U82 4.0V -> J82/J83 pin1.
40mA combined planning load, not a measurement.
Keep the Schmitt buffer and pullups at3.3V; signal inputs tolerate up to5.5V.
C804/C805 are polarized: positive pad1 goes to the named supply.

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication
**Team 4**
Sheet: /Rear zone/
File: Rear.kicad_sch
**Title: Rear zone — integrated logic and interfaces**
Size: A0
Date: 2026-10-23
**Rev: 0.3**
KiCad E.D.A. 10.0.4
Id: 3/5

# Cockpit zone — dual CAN

## CAN A: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

## CAN B: 16 MHz / 500 kbit/s / 3.3 V

RS = GND: high-speed mode. Termination remains on BUS side.
No SPI or clock net is shared between zones. Separate crystal per controller.

## RASPBERRY PI 4 — socket covers physical pins3–38

Rows 9–13 intentionally unused on Pi (Lab3 p17).
Two GND pins per probe group. Active-high LED outputs.
Physical reachability is a layout check, not established by this schematic.

Pi pin4 receives board5.1V. Socket omits1/2/39/40; Pi3.3V pin17 is NOT joined to board3.3V.
Do not attach independent USB-C power while board 5.1V is connected.
Align first socket pair to Pi3/4, not1/2. Check mating unpowered during assembly.

Local bypass is not a load-step proof. Pi bulk capacitance / rail transient test remains open.

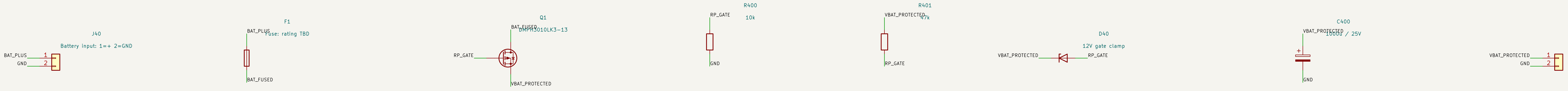
All Pi GPIOs are 3.3V signals. Header pin number is NOT BCM GPIO number.
Independent board 3.3V supplies the six CAN interfaces.
A=CE0/IRQ25; B=CE1/IRQ24; confirm Linux labels against the installed overlay.

Shared rails and CAN backbone connect through the root sheet. SPI, clocks and reset stay within each zone.

Engineering review; actuator/protection and mechanical sign-off pending.
PRELIMINARY — incomplete board; not for fabrication
**Team 4**
Sheet: /Cockpit zone/
File: Cockpit.kicad_sch
**Title: Cockpit zone — integrated logic and interfaces**
Size: A1
Date: 2026-10-23
**Rev: 0.3**
KiCad E.D.A. 10.0.4
Id: 4/5

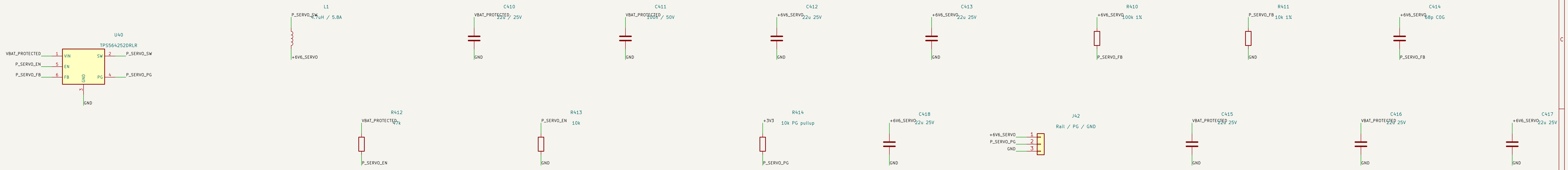
## 3S LiPo INPUT — CNHL S220403BK / XT60; external Hitec 44295 balance charger

Selected pack: 11.1 V nominal / 12.6 V full. Circuit review envelope: 8–15 V.
Battery/motor rail is UNREGULATED. Pack /11 monitor on Front PA3; firmware cutoff must be integrated.
Disconnect and balance—charge off the car. No onboard lithium charger, BMS or hard undervoltage disconnect.



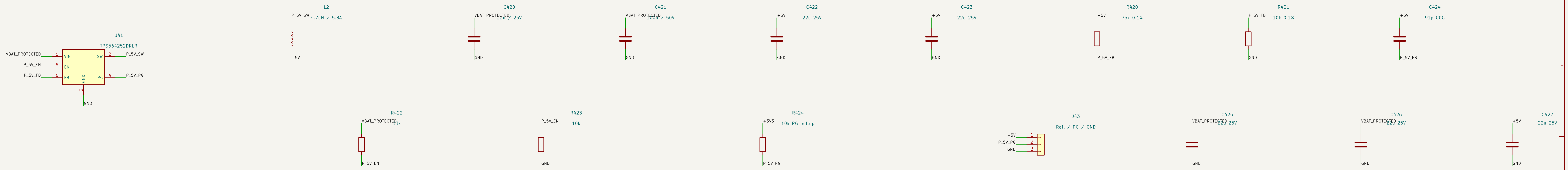
Q1 loss at 10 A: 1 W using 10 mOhm at 25 C; hotter silicon has higher resistance.
F1/holder/cable limits, transient overvoltage and thermal layout are open release checks.
J40 is Hirose DF63M—2P—3.96DSA; verify mating contacts, wire gauge, polarity and fuse coordination.

## SERVO: 6.6 V; allocation 4 A; TPS564252 (4 A device rating)



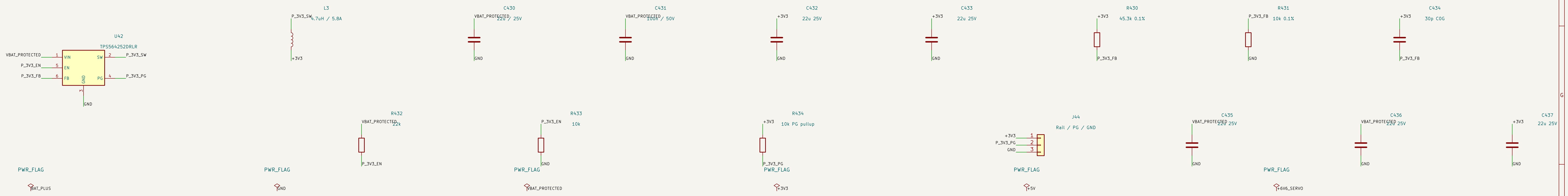
3 x 22uF CIN + 100nF; 4 x 22uF COUT, GRM32ER61E226KE15L (25V/1210).
DC—bias/tolerance/temperature/aging sensitivities documented; loop/transient/thermal behavior unmeasured.

## 5V: 5.1 V; allocation 3.3 A; TPS564252 (4 A device rating)

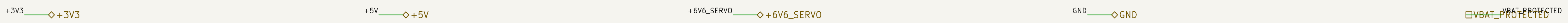


3 x 22uF CIN + 100nF; 3 x 22uF COUT, GRM32ER61E226KE15L (25V/1210).
DC—bias/tolerance/temperature/aging sensitivities documented; loop/transient/thermal behavior unmeasured.

## 3V3: 3.318 V; allocation 0.8 A; TPS564252 (4 A device rating)



3 x 22uF CIN + 100nF; 3 x 22uF COUT, GRM32ER61E226KE15L (25V/1210).
DC—bias/tolerance/temperature/aging sensitivities documented; loop/transient/thermal behavior unmeasured.



Engineering review; actuator/protection and mechanical sign—off pending.
PRELIMINARY — incomplete board; not for fabrication
**Team 4**
Sheet: /Power/
File: Power.kicad_sch
***Title: Power — 3S LiPo input and three buck converters***
Size: A2 Date: 2026—10—23 **Rev: 0.3**
KiCad E.D.A. 10.0.4 Id: 5/5